

# Utility Theory & Decision-Theoretic Expert Systems

## 1. Utility Theory

### 📌 Definition

**Utility Theory** is a concept used in AI to represent and measure **preferences of an agent** over different outcomes.

👉 It helps answer:

*“Which action should I choose when outcomes are uncertain?”*

### 📌 Key Idea

- Instead of just choosing a *likely* outcome, AI chooses the **most preferred outcome**.
- Each outcome is assigned a **utility value (numerical score)**.

### 📌 Example (Simple)

Suppose a student must choose between:

| Option      | Outcome    | Utility |
|-------------|------------|---------|
| Study       | Good marks | 90      |
| Watch Movie | Enjoyment  | 60      |

👉 AI agent chooses **Study** because utility is higher.

### 📌 With Uncertainty

Suppose:

| Action | Outcome | Probability | Utility |
|--------|---------|-------------|---------|
| Study  | Pass    | 0.9         | 100     |
| Study  | Fail    | 0.1         | 0       |
| Movie  | Pass    | 0.5         | 100     |
| Movie  | Fail    | 0.5         | 0       |

👉 Compute **Expected Utility (EU)**:

- $EU(\text{Study}) = (0.9 \times 100) + (0.1 \times 0) = \mathbf{90}$

- $EU(\text{Movie}) = (0.5 \times 100) + (0.5 \times 0) = 50$

✓ Best choice = **Study**

### 📌 Importance

- Used in **decision making under uncertainty**
- Helps AI systems behave **rationally**
- Applied in:
  - Robotics
  - Medical diagnosis
  - Game playing

## 2. Utility Functions

### 📌 Definition

A **Utility Function** maps:

[  
 $\text{Outcome} \rightarrow \text{Utility Value}$   
 ]

👉 It converts real-world results into numbers representing satisfaction.

### 📌 Types of Utility Functions

#### 1. Linear Utility

- Direct proportional relationship

Example:

[  
 $U(x) = x$   
 ]

✓ More profit → more utility

#### 2. Risk-Averse Utility (Concave)

- Prefers **safe outcomes**

Example:

[  
 $U(x) = \sqrt{x}$   
 ]

- ✓ Small guaranteed reward preferred over risky large reward

### 3. Risk-Seeking Utility (Convex)

- Prefers **risky outcomes**

Example:

$$[ \\ U(x) = x^2 \\ ]$$

- ✓ Willing to take risks for higher rewards

#### **Example**

A person chooses between:

| Option | Outcome                        |
|--------|--------------------------------|
| A      | ₹100 sure                      |
| B      | 50% chance ₹200, 50% chance ₹0 |

#### Case 1: Linear Utility

- Both equal → indifferent

#### Case 2: Risk-Averse

- Prefers **Option A**

#### Case 3: Risk-Seeking

- Prefers **Option B**

#### **Graphical Understanding (conceptual)**

- Concave → Risk-averse
- Convex → Risk-seeking
- Straight line → Risk-neutral

## 📌 Properties of Utility Functions

- Higher value = more preference
- Can handle **uncertainty + trade-offs**
- Used with probability → Expected Utility

## 3. Decision-Theoretic Expert Systems

### 📌 Definition

A **Decision-Theoretic Expert System** is an AI system that:

- ✓ Combines **probability theory + utility theory**
- ✓ Makes decisions under uncertainty
- ✓ Chooses action with **maximum expected utility**

### 📌 Traditional vs Decision-Theoretic

| Feature     | Traditional Expert System | Decision-Theoretic |
|-------------|---------------------------|--------------------|
| Logic       | Rule-based                | Probabilistic      |
| Uncertainty | Not handled well          | Handled            |
| Decision    | Fixed rules               | Utility-based      |

### 📌 Components

1. **Knowledge Base**
  - Facts and rules
2. **Probability Model**
  - Likelihood of events
3. **Utility Function**
  - Preferences of outcomes
4. **Inference Engine**
  - Computes best decision

### 📌 Example: Medical Diagnosis System

A doctor must decide whether to **treat a patient**.

| Action | Outcome | Probability | Utility |
|--------|---------|-------------|---------|
| Treat  | Cure    | 0.8         | +100    |

| Action   | Outcome      | Probability | Utility |
|----------|--------------|-------------|---------|
| Treat    | Side Effects | 0.2         | -20     |
| No Treat | Cure         | 0.3         | +100    |
| No Treat | Disease      | 0.7         | -100    |

### ✚ Expected Utility Calculation

- $EU(\text{Treat}) = (0.8 \times 100) + (0.2 \times -20) = 80 - 4 = \mathbf{76}$
- $EU(\text{No Treat}) = (0.3 \times 100) + (0.7 \times -100) = 30 - 70 = \mathbf{-40}$

✓ Best Decision → **Treat**

### ✚ Applications

- Medical diagnosis systems
- Financial decision systems
- Autonomous vehicles
- Business decision support

### ✚ Advantages

- ✓ Handles uncertainty effectively
- ✓ More realistic decision-making
- ✓ Flexible and adaptive

### ✚ Limitations

- ✗ Requires accurate probabilities
- ✗ Difficult to design utility functions
- ✗ Computational complexity

## Summary

| Concept          | Key Idea                    |
|------------------|-----------------------------|
| Utility Theory   | Measures preference         |
| Utility Function | Maps outcome → satisfaction |

| Concept                   | Key Idea                                   |
|---------------------------|--------------------------------------------|
| Decision-Theoretic System | Chooses action using probability + utility |